causalgeo: channel effects from concurrent geo holdouts

Aditya Puttaparthi Tirumala[a,*]

[a]*Independent Researcher*

Abstract

Marketing teams sometimes turn off different bundles of media channels in different groups of markets at the same time. These geo holdouts measure each bundle's incremental effect, but not how that effect should be divided among its channels. Existing geo tools stop at the bundle. Marketing-mix models can split it, but the answer is unstable when channels move together.

`causalgeo` assigns markets to concurrent cells, or takes an assignment the user already has. It estimates each bundle from the holdout, checks whether a channel split is supported, pulls channel slopes toward the measured totals, reports the gap, and withholds slopes the data cannot separate. One constructed series recovers the slopes within 0.21 percent. That series is the linear model the package fits. Across 200 draws the median error is 0.19 percent, and 0.76 percent at four times the noise, with coverage staying at 95.8 percent. A confounder the holdout does not turn off leaves a slope 50 percent high. When two channels are nearly copies and the effects are small, calibration lowers the median error from 44.6 percent to 10.5 percent. Omitted carryover or saturation still breaks the linear split.

Keywords: geo experiment, incrementality, holdout, marketing mix, Python

1. Motivation and significance

Geo experiments estimate incremental impact by comparing markets assigned to a media change with similar control markets [1, 2]. This avoids the bias in comparing people who saw an ad with people who did not [3, 4]. It answers a simple question: did the channel create incremental sales? [5].

*Corresponding author.

Email address: `puttaparthy.aditya@gmail.com` (Aditya Puttaparthi Tirumala)

Table 1: Code metadata.

Nr.	Code metadata description	Value
C1	Current code version	0.2.0
C2	Permanent link to code/repository used for this code version	`https://github.com/adityapt/causalgeo/releases/tag/v0.2.0`
C3	Permanent link to Reproducible Capsule	`https://github.com/adityapt/causalgeo/blob/v0.2.0/paper/reproduce.py`
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python
C7	Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10. Runtime dependencies: pandas, numpy, scipy, statsmodels. The example also uses matplotlib.
C8	Link to developer documentation	`https://github.com/adityapt/causalgeo/blob/main/README.md`
C9	Support email for questions	puttaparthy.aditya@gmail.com

The number of markets that can go dark is usually limited. Consider a United States campaign across 210 designated market areas (DMAs). The business may allow a pause in only 30: large markets may be too risky, some regions may have partner commitments, and others may already be in a campaign. Those business rules set the holdout pool before the analyst can consider balance or power. Advertising effects are small compared with sales noise [6], so one well-powered single-channel test can use the whole pool. Teams therefore test bundles in several cells at once. Each cell identifies a bundle effect, not the effects of the channels inside it.

GeoLift and Meridian GeoX can design and measure more than one cell [7, 8]. They do not split a cell into channels or tie a mix model to the measured cell total. Mix models estimate channel slopes from impression histories, and collinear media makes that split arbitrary [9, 10]. Robyn and PyMC-Marketing can pull a mix model toward a lift test, including a test of

more than one channel [11, 12]. A pull toward an experiment can stabilize a split [13]. It cannot create information the test and the exposure data do not contain. The split below also assumes that a bundle effect is the sum of its channel effects. If channels interact, these equations do not name the separate slopes [14].

`causalgeo` bridges the experiment and the channel model. It measures the bundle effect from the holdout, then uses that measurement to guide a linear model of channel impressions. The resulting channel slopes use both the experiment and the model; they are not direct experimental estimates. If the data cannot separate channels, the package reports the bundle instead. It does not control for test-period spend because the holdout itself changes spend [15].

2. Software description

2.1. Software architecture

The concurrent workflow, exposed through `design_concurrent_test` and `disentangle`, has four parts.

1. `design.py` assigns markets within strata, builds the cell-by-channel schedule, and reports which channels are separable.
2. `identified.py` estimates one bundle effect per cell and reports a direct channel effect only when that channel was tested alone.
3. `exposure.py` checks whether channels that should stay on changed differently in test and control markets.
4. `calibrated.py` fits the impression model and uses each measured bundle effect to guide the channel split.

Bundle effects use market and week fixed effects. Market and week means are removed iteratively, so a market that misses some weeks is handled in the same way as market and week dummies. Standard errors are clustered by market [16, 17, 18]. This example has 40 markets and 8 treated markets in each cell. On a separate panel with those sizes and 30 periods, redrawing the assignment 200 times, the nominal 95 percent intervals covered the true cell effect 91.8 percent of the time. The supporting `analyze` function also provides time-based regression [19], synthetic control [20, 21], and synthetic difference-in-differences [22] for a single-cell test that has already been assigned. GeoLift and GeoX handle market selection and power search [7, 8].

2.2. Software functionalities

Design. The user supplies markets, strata, channels, and the channels to turn off in each cell. Within each stratum, `control_share` remains on; the other markets are divided among cells. `design_from_assignment` accepts an assignment that already exists, with the same cell labels. The schedule has $A_{kc} = 1$ when cell k turns channel c off. The separation report calls a channel not tested, tied, separate, or not separable. A channel the design never turns off stays `not tested` and receives no slope.

Bundle effect. Let $D_{it}^{(k)} = 1$ when market i is assigned to cell k and week t is in the test window. The cell regression is

$$Y_{it} = \sum_k \tau_k \, D_{it}^{(k)} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

τ_k is the cell's sales change after allowing for each market's history and common week shocks. It is a channel effect only when one channel was tested alone. Even if overlapping bundles cover all channels, a bundle coefficient does not belong to any one channel.

Still-on channels. A channel left on in cell k can still differ between the dark markets and the control. The impression shift is the difference in differences

$$S_{kc} = \big(\bar{X}_{c,k,\text{post}} - \bar{X}_{c,k,\text{pre}}\big) - \big(\bar{X}_{c,\text{control,post}} - \bar{X}_{c,\text{control,pre}}\big). \tag{2}$$

The pre-test slope b_c is the joint slope of all channels, not a regression of sales on one channel. The usual cell estimate stays τ_k. If $|b_c S_{kc}|$ is more than `material` times the larger of $|\tau_k|$ and the pre-test outcome standard deviation, with `material` equal to 0.10 unless the user changes it, the channel split uses a refit that includes these still-on impressions. The user can also choose to report that refit. Turned-off channels are never controls: being off is the treatment. The example table reports the requested refit.

Channel split. Let W_{kc} be the impressions of channel c withheld in cell k,

$$W_{kc} = \max(-S_{kc},\, 0) \tag{3}$$

for channels that cell k turned off; all other entries in that calibration row are zero. The model should make $\sum_c \beta_c W_{kc}$ close to $-\tau_k$. This is a weighted observation, not a hard constraint. The impression model uses all pre-test

observations and only control markets during the test; dark-market weeks are excluded. Precise bundle estimates receive more weight. The bundle table shows any remaining gap.

The reported covariance is the inverse weighted cross-product. It treats bundle estimates as independent, so it does not fully account for shared controls or reused observations. The simulations assess this approximation for this example design, not every possible panel.

Columns with cosine similarity above $1 - \texttt{copy_tolerance}$ are treated as copies. The default tolerance is 10^{-6}. One copy stays in the model to account for their shared signal, but neither gets a slope. By default, a slope is marked unstable if it changes substantially when the calibration weight is reduced or when the market with the largest test-period outcome is removed. The tables below turn that check off, so the reported numbers are the point estimates. A blank impression stays blank, that market-day is dropped, and the fit warns. Daily impact is exposure times $\hat{\beta}_c$. The package offers geometric adstock but not saturation because one on/off holdout cannot trace a response curve [23, 10].

The output includes channel slopes, measured and model-implied bundle effects, the exposure check, and daily impacts. A market-cluster bootstrap reruns the pipeline on markets resampled within cells.

3. Illustrative examples

No public geo holdout ships with known channel-level effects. The example uses the published Meridian geo impression file `geo_all_channels.csv`: 40 markets and 156 weeks, from 25 January 2021 through 15 January 2024 [24]. Sales are constructed so the slopes are known. Meridian's conversions column is used only to form strata. It is not the outcome, because that series already contains Meridian's own media response.

Three of the five channels are used, with planted slopes 2×10^{-4}, 5×10^{-4}, and 1×10^{-4} sales per impression. Strata are quartiles of each market's mean conversions. With `control_share = 0.4` and seed 1, the design places 16 markets in control and 8 in each cell. The test window is the last 16 weeks, beginning 2 October 2023. In a dark market-week the turned-off channel's impressions are set to zero, and sales are

$$Y_{it} = \alpha_i + \gamma_t + \sum_c \beta_c X_{itc} + \varepsilon_{it}, \tag{4}$$

Table 2: Cell effects on the constructed series, with the still-on refit requested. The default leaves the unadjusted column in place and only records the shift. The model column values the off-channel impression change at the fitted slopes.

Cell	Bundle	Unadjusted	Adjusted	SE	Model
1	Channel0+Channel1	−555.7	−555.7	62.9	−570.2
2	Channel0+Channel2	−318.0	−251.5	36.2	−253.6
3	Channel1+Channel2	−399.9	−399.9	54.4	−396.2

Table 3: Planted and recovered slopes, in sales per 10,000 impressions. The largest absolute error is 0.21 percent.

Channel	Planted	Recovered	SE	Abs. error (%)
Channel0	2.000	1.997	0.006	0.17
Channel1	5.000	5.009	0.008	0.17
Channel2	1.000	1.002	0.008	0.21

with $\alpha_i \sim N(0, 50^2)$, $\gamma_t \sim N(0, 20^2)$, and $\varepsilon_{it} \sim N(0, 30^2)$, drawn with seed 7. The script is `paper/example.py`.

The cells turn off {Channel0, Channel1}, {Channel0, Channel2}, and {Channel1, Channel2}, yielding rows $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. The schedule has full column rank, but no channel is tested alone; the assignment-only table therefore leaves all three channel effects blank. Table 3 instead reports the impression-model slopes, softly calibrated to the three measured bundles.

This is a recovery check for the linear model that the package fits, not evidence about saturation, long carryover, or the power of an eight-market cell. The impression paths are external to the package; sales are constructed from known slopes. Real experiments with smaller effects will have wider intervals [6].

Cell 2 illustrates the exposure correction. Channel1's still-on shift, valued at the joint pre-test slope, is worth −59 sales, or 19 percent of the unadjusted effect, and refitting moves the estimate from −318.0 to −251.5. Cells 1 and 3 stay at the unadjusted experiment.

Table 2 compares experimental and model-implied bundle effects. Their gaps are 2.6, 0.8, and 0.9 percent of the experimental effects; all model totals lie within the experimental intervals. The bundle coefficient remains the

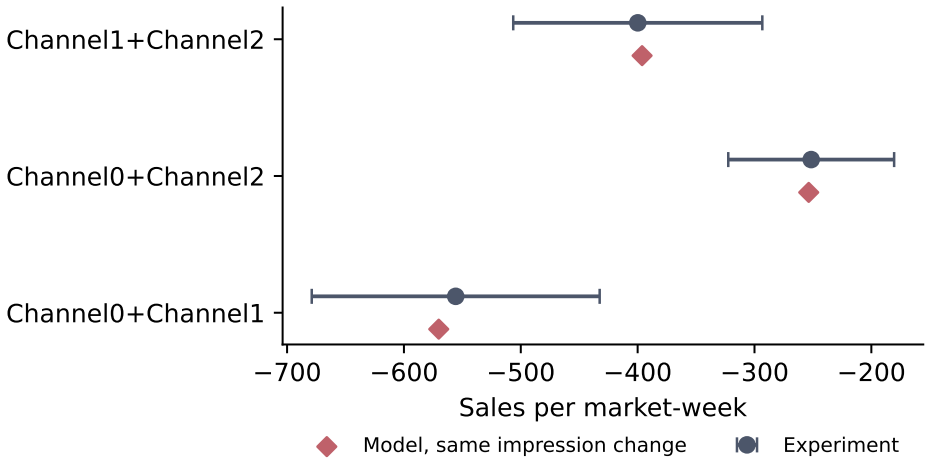


Figure 1: Experimental bundle effect, with a 95 percent interval, and the same off-channel impression change valued at the fitted slopes. Sales per market-week.

experimental result, while the channel slope is the model-assisted split.

The same three cell totals and the withheld-impression matrix can be solved for the three channel slopes with no observational market-weeks. That experiment-only split gives 2.22, 4.48, and 2.01 sales per 10,000 impressions, with standard errors 0.43, 0.64, and 1.14. The calibrated standard errors on the same series are 0.006, 0.008, and 0.008. The tighter intervals come from the impression histories, not from the three cell totals.

When Channel1 is replaced with a copy of Channel0, both are withheld. Their shared path remains as one control: Channel2 is then recovered as 0.97 sales per 10,000 impressions, versus its planted 1.00. Omitting the shared path instead biases it to 3.02. Figure 2 shows those two fits against the planted value.

Table 4: Two hundred refits per setting. Rho is the mixing weight on Channel0 in the untreated Channel1 path, and the correlation is what that mix produces. Noise multiplies the example residual standard deviation. Error is the median absolute percent error among reported slopes. Coverage brackets are 95 percent Monte Carlo intervals. At rho 1, error and coverage refer to the remaining channel.

Noise	Rho	Corr.	Runs	Error (%)	Coverage [MC interval]	Withheld
1	0.0	0.34	200	0.19	0.958 [0.942, 0.974]	0.00
1	0.9	0.96	200	0.40	0.953 [0.934, 0.973]	0.00
1	1.0	1.00	200	0.58	0.940 [0.907, 0.973]	1.00
4	0.0	0.34	200	0.76	0.958 [0.942, 0.974]	0.00
4	0.9	0.96	200	1.62	0.957 [0.938, 0.975]	0.00
4	1.0	1.00	200	2.25	0.940 [0.907, 0.973]	1.00

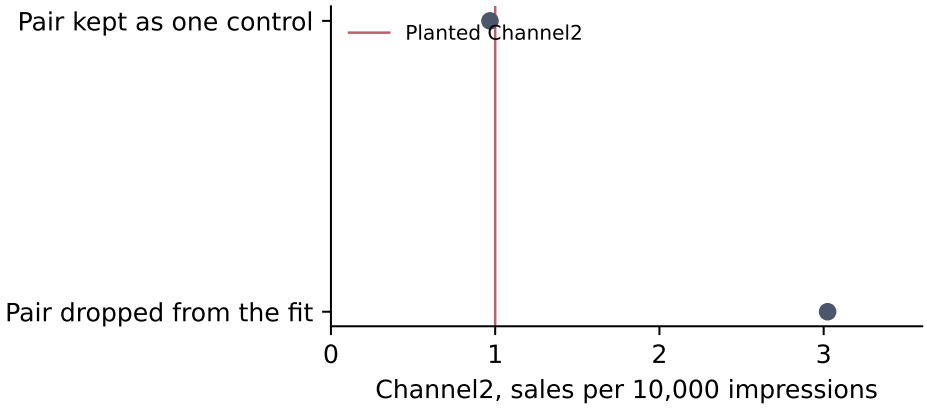


Figure 2: Channel2 when Channel1 is an exact copy of Channel0. Keeping the shared path as one control recovers the planted slope. Dropping that path pushes the remaining slope from about 1 to about 3 sales per 10,000 impressions.

Table 4 and Figure 3 summarize 200 draws for each mixing weight and noise level. At the example noise, median absolute error is 0.19 percent when the paths are distinct, where their correlation is 0.34, and 0.40 percent at mixing weight 0.9, where the correlation is 0.96. Coverage is 95.8 and 95.3 percent. At an exact copy, the pair is withheld in every run and the remaining channel covers in 94.0 percent of runs. Multiplying the residual noise by four raises the distinct-path error from 0.19 to 0.76 percent and leaves coverage at 95.8 percent. These results assess this design and the diagonal covariance approximation.

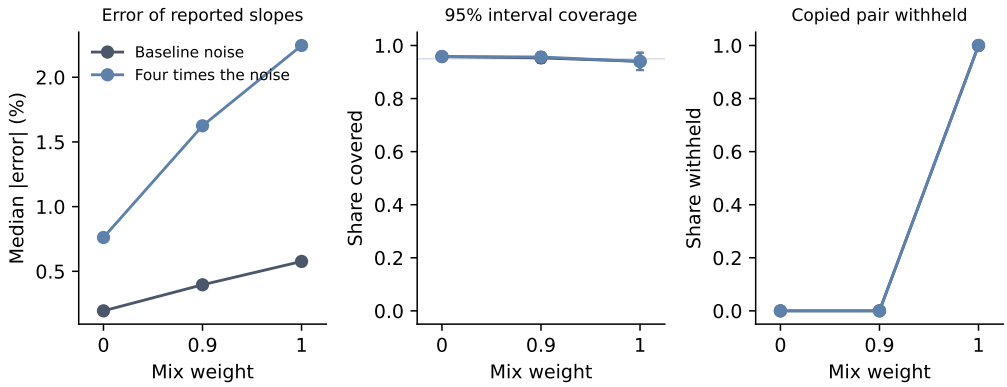


Figure 3: Recovery over 200 simulations as the untreated Channel1 path is mixed toward Channel0. The second series multiplies residual noise by four. Bars in the coverage panel are 95 percent Monte Carlo intervals. At mix weight 1 the pair is withheld, so error and coverage refer to the remaining channel.

Table 5: Misspecification stress test over 200 draws. Error is the median absolute percent error in channel contribution to the experimental contrast. Coverage is the mean coverage of its nominal 95 percent interval; brackets give the 95 percent Monte Carlo interval. Gap is the median absolute model–experiment bundle gap as a percentage of the experimental effect.

Fit	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Carryover omitted	200	47.27	0.000 [0.000, 0.000]	48.09
Carryover modeled	200	0.19	0.942 [0.922, 0.961]	2.22
Saturation omitted	200	26.99	0.000 [0.000, 0.000]	26.97

The preceding recovery check uses the model that generated sales. Table 5 instead evaluates two omitted response mechanisms over 200 draws. Geometric carryover has decay 0.5. The saturation response is $\beta_c X/(1+X/h_c)$, where h_c is twice the median positive exposure and β_c is the slope at zero. Because neither process has one true raw-impression slope, error and coverage refer to each channel's known contribution to the same pre/post, test/control contrast as the experiment. The carryover model is also refit with the planted decay to check the package's optional adstock transformation.

Supplying the planted carryover decay restores the recovery seen under the linear model. Omitting carryover or saturation produces large errors and no interval coverage. The corresponding bundle gaps are also large, so

the package exposes the disagreement rather than making the misspecified channel split look experimentally validated. These stress tests do not select a decay or estimate a saturation curve.

Table 6: Further checks, 30 draws each, so the coverage intervals are wide. Error is the median absolute percent error among reported slopes. The confounded row instead reports the Channel0 error. Coverage brackets are 95 percent Monte Carlo intervals. Serial correlation adds an AR(1) residual with autocorrelation 0.8. Heterogeneous slopes vary by market between half and one and a half times the planted slope. Redistribution moves half of a typical withheld week onto a channel the cell left on. Incomplete delivery leaves 30 percent of the control level running. The confounder follows Channel0's business-as-usual path and does not turn off. The collinear rows mix Channel1 toward Channel0 with weight 0.999, correlation 0.9995, and use slopes at 5 percent of the example.

Setting	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Serial correlation	30	0.43	0.956 [0.914, 0.997]	3.56
Heterogeneous slopes	30	4.81	0.344 [0.248, 0.441]	4.01
Budget redistributed	30	0.22	0.967 [0.930, 1.000]	12.60
Redistribution, refit requested	30	0.22	0.967 [0.930, 1.000]	12.60
Incomplete delivery	30	0.22	0.967 [0.930, 1.000]	2.83
Observational fit only	30	0.22	0.967 [0.930, 1.000]	2.12
Calibrated fit	30	0.22	0.967 [0.930, 1.000]	2.13
Confounded Channel0	30	49.94	0.644 [0.614, 0.675]	21.12
Collinear, observational	30	44.64	0.956 [0.895, 1.000]	43.33
Collinear, calibrated	30	10.52	0.989 [0.967, 1.000]	8.79
Experiment only	30	11.16	1.000 [1.000, 1.000]	–

Table 6 asks what the linear recovery depends on. Serial correlation and a recorded shortfall in delivery leave the channel error near the linear result. Market-specific slopes do not: coverage falls to 34.4 percent. Moving the lost impressions onto a still-on channel leaves the channel error near 0.2 percent. The raw cell total sits 52.9 percent away from the total after those extra impressions are removed, because the experiment contains the reallocated delivery. The channel split is calibrated to the cleaned total, and the remaining bundle gap is 12.6 percent. Replacing the reported cell estimate with that refit does not change the split, so the two budget rows agree. On the original series the observational fit and the calibrated fit both have 0.22 percent error. The calibration rows are not what produces that recovery.

A confounder that tracks Channel0 and stays on during the holdout leaves that slope 49.9 percent high after calibration, against 50.0 percent with the

observational fit. The bundle gap opens to 21 percent. The diagnostic moves. The slope does not. Calibration does change the split when the channels are nearly copies and the planted slopes are small. At mixing weight 0.999 the observational median error is 44.6 percent and the calibrated error is 10.5 percent, and the bundle gap falls from 43 percent to 8.8 percent. Both of those coverage intervals come from 30 draws and are wide: the observational interval runs from 0.895 to 1, and the calibrated interval runs from 0.967 to 1. The experiment-only split, refit on the original linear series, has median error 11.2 percent and coverage 1 because its intervals are wide. Figure 4 puts the coverage and gap pattern next to the omitted-response checks.

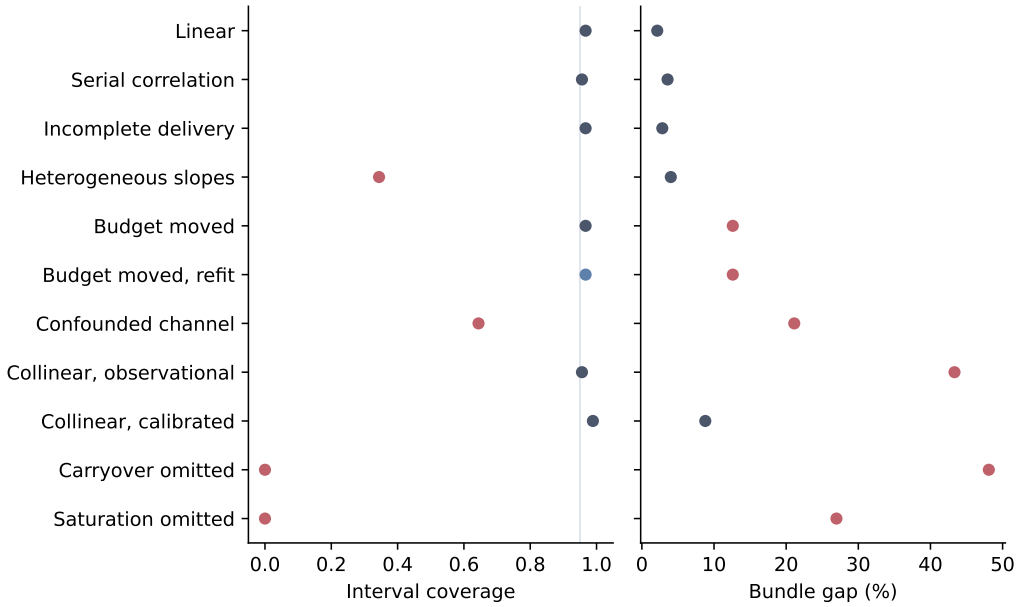


Figure 4: Coverage and the bundle gap. The linear, serial, delivery, slope, budget, confounder, and collinear rows are 30 draws. Carryover and saturation are the 200-draw checks in Table 5. The pale line is 95 percent coverage.

4. Impact

The package gives teams one place to assign cells, measure bundle effects, check which channels can be separated, and report model-assisted channel slopes next to the remaining model–experiment gap. GeoLift and GeoX can run more than one cell; they do not make this split [7, 8]. Robyn and

PyMC-Marketing already calibrate a mix model to lift tests [11, 12]. This package is the concurrent holdout, the reported gap, and the refusal when channels cannot be separated. The bundle is the directly measured result. The channel is the finer, model-assisted result.

The current release is 0.2.0. Adoption is early, and this paper has no user study. The reusable example, `paper/reproduce.py`, rebuilds all five tables and four figures. The same functions apply to a real holdout once the user supplies market outcomes and channel impressions. Channels the design cannot separate, or whose impressions are copies, remain blank rather than receiving a made-up slope.

5. Conclusions

`causalgeo` measures concurrent channel-bundle holdouts and splits a measured bundle only when the impression data support it. The calibration is soft: it guides the channel model toward the experiment rather than forcing agreement. On one constructed series the slopes are within 0.21 percent of the planted values. Solving the same cell totals with no impression model gives standard errors of 0.43 to 1.14 sales per 10,000 impressions, against 0.006 to 0.008 for the calibrated split. Across 200 draws, four times the residual noise raises median error from 0.19 to 0.76 percent and leaves coverage at 95.8 percent. Market-specific slopes cut coverage to 34.4 percent. A confounder the holdout leaves on keeps the affected slope about 50 percent high, while nearly copied channels with small effects move from 44.6 percent error to 10.5 percent once the cell totals are included. A matched carryover fit gives 94.2 percent coverage. Omitted carryover or saturation produces large channel errors and bundle gaps. On a 30-period panel with the same market counts, cell intervals covered the true effect in 91.8 percent of 200 re-randomized assignments. Intervals remain conditional on the response model and the diagonal covariance approximation.

The split is linear, with optional geometric adstock and no saturation curve [10]. A channel being on or off does not reveal the full response curve, so the package does not estimate one, and it does not estimate an interaction between channels [14]. GeoLift and GeoX handle market selection and power [7, 8]. This package assigns markets within strata the user has already chosen, or accepts that assignment, and does not claim that eight markets make a powered cell [6]. The published fits leave the stability check off. The default cell result is the unadjusted experiment. When lost impressions move to

a channel left on, the split uses the refitted cell total after that delivery is removed. Test-period spend stays out of the regression because assignment changes it [15, 3].

CRediT authorship contribution statement

Aditya Puttaparthi Tirumala: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The impression paths are the Meridian file `geo_all_channels.csv` [24]. `examples/meridian_conversions.py` runs the same split on Meridian's conversions column. That run has no known channel effects. It only shows the software on that file. In `paper/`, set `PYTHONPATH=../src` and run `python3.11 reproduce.py`. That rebuilds Tables 2–6 and the four figures, then checks them against the rounded values reported here. `python3.11 reproduce.py --fast` uses two draws per setting and only checks that the scripts finish. Table 1 gives the repository links.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor, with the Grok language model, to draft and revise the manuscript and the example code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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